

[HCI/2024/Q1]

1 Name your Jupyter Notebook as

```
TASK1_<your name>_<centre number>_<index number>.ipynb
```

The task is to implement columnar transposition cipher to encode a message.

The columnar transposition cipher is a type of transposition cipher in which the plain text is written into a grid of rows and columns based on a keyword. The cipher text is formed by reading the characters off the grid column by column in the alphabetical order of the keyword's letters. This method rearranges the characters of the plaintext, making it difficult to decipher without knowing the specific key and the order of rearrangement.

The following steps describe how columnar transposition is used to encode a plain text to a cipher text.

- All spaces between each word in the plain text are replaced by 'z'.
- Write the plain text in rows of length equal to the number of letters in the keyword in a grid. If the plain text does not fit perfectly in the row of the grid, fill the empty spaces with 'x'.

For example, if the plaintext is "computing is an interesting subject", and the keyword is "ALGORITHM", the grid will look as follows:

A	L	G	O	R	I	T	H	M
c	o	m	p	u	t	i	n	g
z	i	s	z	a	n	z	i	n
t	e	r	e	s	t	i	n	g
z	s	u	b	j	e	c	t	x

- Assign a number to each letter of the keyword in alphabetical order. The cipher text is obtained by reading down the columns in ascending order of the numbers.

1	5	2	7	8	4	9	3	6
A	L	G	O	R	I	T	H	M
c	o	m	p	u	t	i	n	g
z	i	s	z	a	n	z	i	n
t	e	r	e	s	t	i	n	g
z	s	u	b	j	e	c	t	x

Cipher text: cztzmsruninttnteoiesgngxpzebuasjizic

[HCI/2024/Q1 Cont']

For each of the sub-tasks, add a comment statement at the beginning of the code using the hash symbol '#', to indicate the sub-task the program code belongs to, for example:

In [1]:

```
#Task 1.1  
Program Code
```

Output:

Task 1.1

Write a function `task1_1(key, message)` that takes in two arguments. The function uses columnar transposition to encode the message by using the key and returns a cipher message. [11]

Task 1.2

Write program code to encode the message "computing is an interesting subject" with key "ALGORITHM" and print the cipher message. [1]

Save your Jupyter Notebook for Task 1.